

Reading and Writing Files in Python

A beginner friendly guide to opening, reading, writing, and appending text files with `with open()`.



1. What file handling means

File handling lets a Python program work with data stored in external files such as notes, logs, reports, or configuration files. For basic automation, text files are the easiest place to start because they store plain readable content.

- A `.txt` file stores plain text.
- Python can **open**, **read**, **write**, and **append** to it.
- The safest beginner pattern is **with open()** because it automatically closes the file.

2. Using with open()

The **with open()** context manager opens the file for a short block of code and then closes it automatically, even if an error happens. This makes the code cleaner and reduces mistakes like leaving files open.

- `open("notes.txt", "r")` - open for reading
- `open("notes.txt", "w")` - open for writing and replace old content
- `open("notes.txt", "a")` - open for appending and keep old content

3. Example: read from a text file

Example 1 - Read

```
with open("notes.txt", "r") as file:
    content = file.read()
    print(content)
```

Use mode `"r"` to read existing text. `file.read()` returns the full file content as one string.

4. Example: write a new text file

Example 2 - Write

```
with open("notes.txt", "w") as file:
    file.write("Python automation starts here.")
```

Use mode `"w"` to create a file or overwrite an old one. This is useful when the program should replace the previous content with fresh output.

5. Example: append more text

Example 3 - Append

```
with open("notes.txt", "a") as file:  
    file.write("\nNew line added later.")
```

Use mode **"a"** when you want to keep the existing text and add more at the end. This is common for logs, reminders, and automatic status updates.

6. Visual guide to the modes

Mode	Purpose	What happens?
r	Read	Opens an existing file and reads its content.
w	Write	Creates a new file or replaces everything inside an old file.
a	Append	Adds new text to the end without deleting old content.

7. Best practices

- Use **with open()** so the file closes automatically.
- Use **\n** when you want a new line in the file.
- Choose **w** carefully because it erases old content.
- Use short, clear filenames such as **notes.txt** or **log.txt**.

Quick recap

- Open a file with **with open()**.
- Read with **read()**.
- Write with **write()** in mode **w**.
- Append with **write()** in mode **a**.